

Employee Career Advancement Prediction

Project by algoranger (kelompok 1)

Data Science Batch 58



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Agenda

Employee Career
Advancement Prediction



Introduction
Business Understanding
Problem Statement & Goals
Understanding Dataset
EDA
Data Pre-Processing
Modeling & Evaluation
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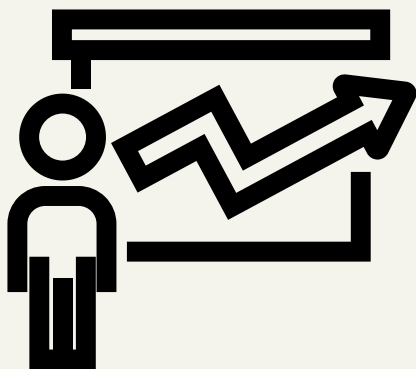
Data Overview

This project focuses on utilizing a Machine Learning model designed to predict employee promotion eligibility by leveraging Employee Career Advancement Prediction Data.

Dataset : Rakamin Bootcamp - Dataset - Promotion Dataset.csv

Info Data :

- Total records: 1,000 employees
- Data missing: 950 (5% missing values)
- Features: 10 columns, 2 numeric and 2 categorical
- Target variable: Promotion_Eligible (1 = eligible, 0 = not eligible)



Head Data :

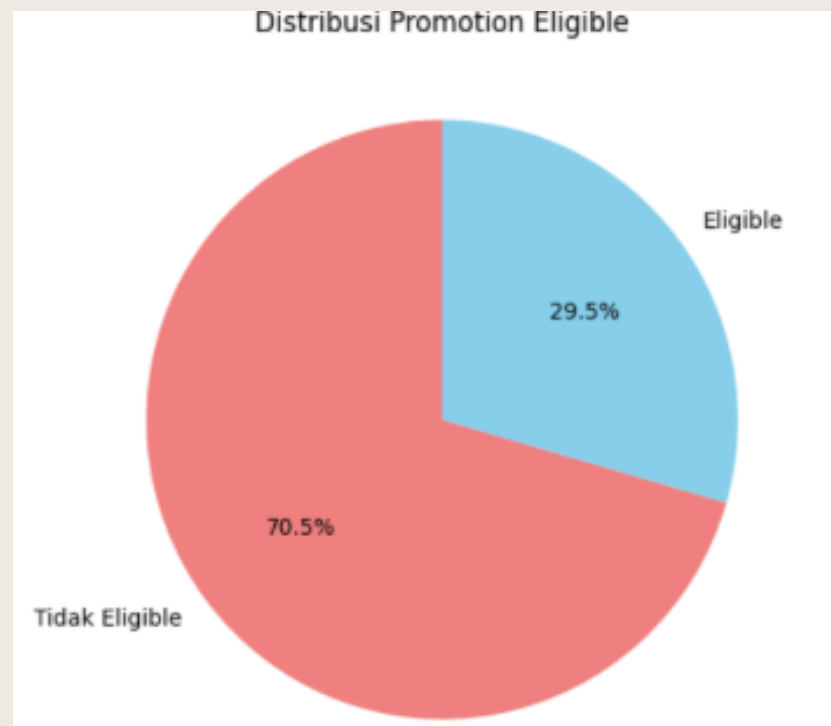
Employee_ID	Age	Years_at_Company	Performance_Score	Leadership_Score	Training_Hours	Projects_Handled	Peer_Review_Score	Current_Position_Level	Promotion_Eligible
EMP0001	24.0	21.0	2.0	44.0	42.0	19.0	61.0	Junior	0.0
EMP0002	50.0	8.0	5.0	45.0	66.0	11.0	6.0	Junior	0.0
EMP0003	56.0	9.0	5.0	19.0	96.0	6.0	86.0	Lead	0.0
EMP0004	39.0	24.0	4.0	51.0	16.0	6.0	52.0	Senior	0.0
EMP0005	41.0	NaN	4.0	64.0	57.0	16.0	59.0	Junior	0.0

Business Understanding: Employee Promotion Process

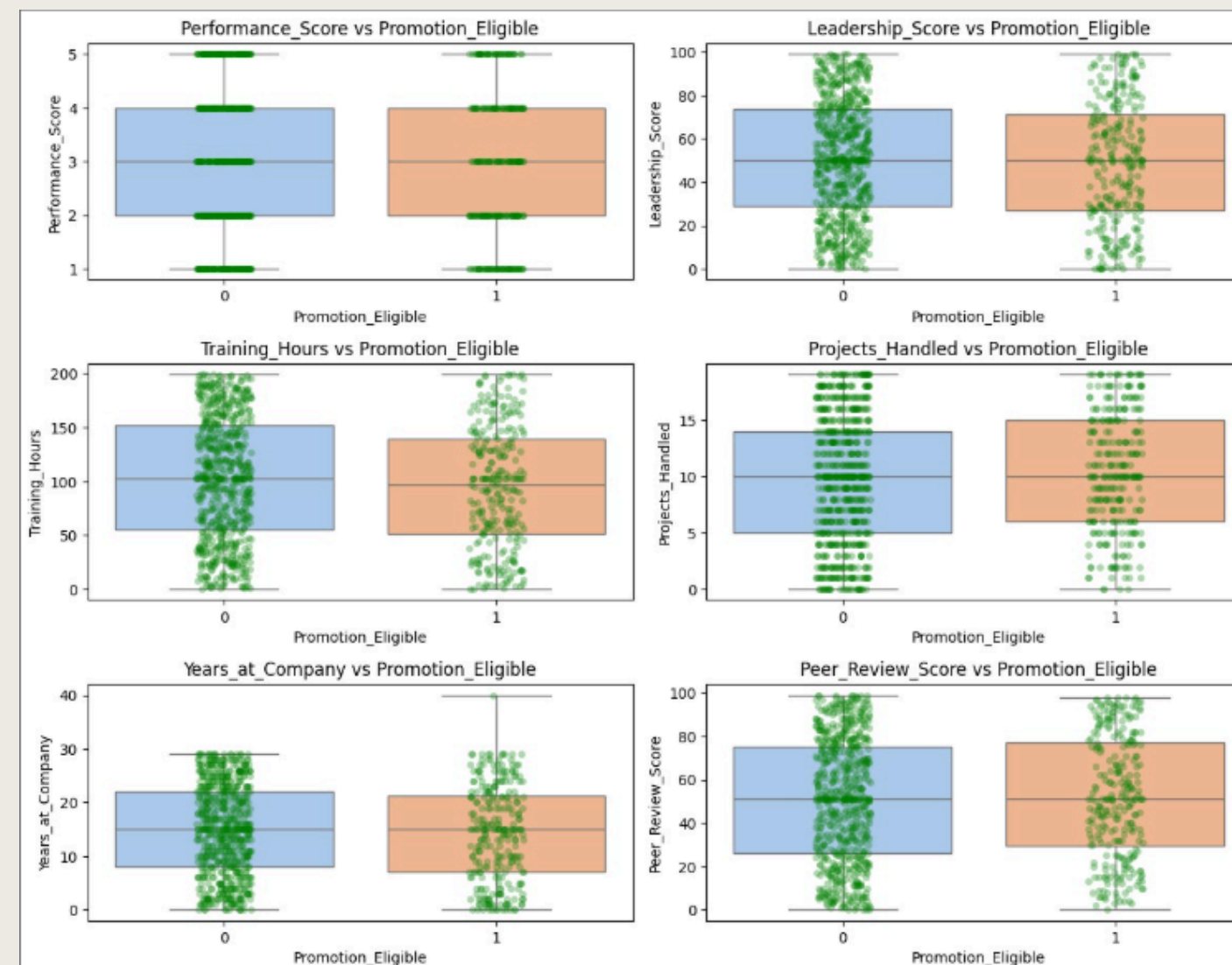
Business Context

- Employee promotion → aims to support business operations and employee development
- Process → multi-step review involving several stakeholders (managers, HR, promotion committee) with various considerations such as performance, age, leadership, projects, and others - multi-factor
- Mechanism → manually

Initial Data Findings



- **29% Eligible**, which is higher than the industry average promotion rate of 8-14% (Mercer Talent Trends & Deloitte, 2023)
- The data indicates a **class imbalance**



The distribution of data for employees who are Eligible (1) and Not Eligible (0) is similar – promotion is not influenced by a single factor → **a non-linear or combination of many factors.**

Supporting Reference

*“Organizations often spend time in ways that don’t increase value — such as slow decision-making, poor prioritization, and low-value meetings.”
(McKinsey & Company, “Increasing Your Return on Talent: The Moves and Metrics That Matter”, 2024)*

Industry Insight

*“By automating manual and repetitive tasks, organizations can reduce administrative and operational costs by 30–50%. ”
(McKinsey & Company, 2020 — “Operations management reshaped by robotic automation”)*



To what extent does the complexity of promotional decision-making impact company operations?

Assumptions::

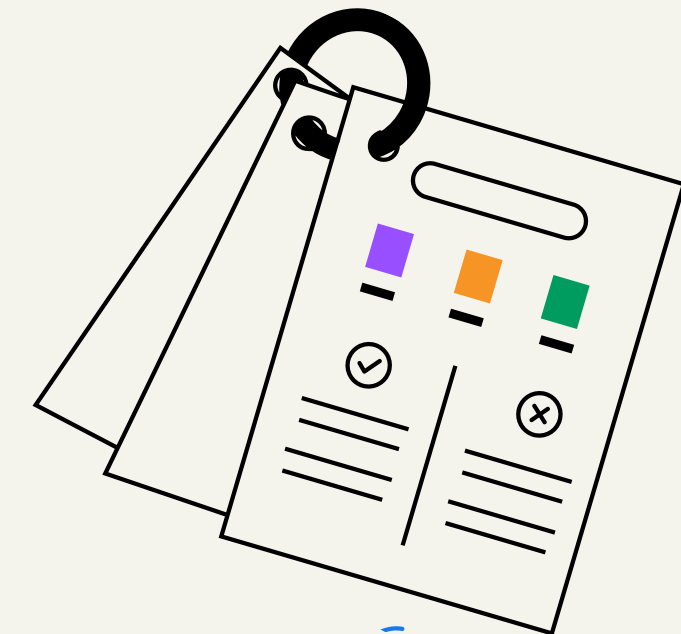
- Company size: 500 employees → 29% eligible (≈145 employees) → Promotion rate: 8–14% or 40–70 promotions per year (ref)
- Review process: 3 reviewers + 1 HR admin per candidate (Schleu & Hüffmeier, 2021; SHRM, 2020)
- Average review time: 1.5 hours per reviewer + 0.5 hour per HR admin
- Average labor cost: USD 40/hour (SHRM Cost-Per-Hire Benchmark, 2020)
- Average decision delay due to process complexity: 14 days (Forbes & HBR, 2019)
- Opportunity cost per day: USD 200 (based on salary and productivity benchmarks)

Manual scheme:

Manual: 500 karyawan → Screening → 145 eligible → Review → 40 promoted

Estimated Costs:

Component	Formula	Cost/ Candidate (USD)	Total Cost (USD)	Total Time (hours)
Initial screening – 500 employees, 1 manager per 50 employees	10 managers x 0.5 hours x 40 USD	200	10000	20
Formal review - 145 employees	4 reviewer x 0.5 hours x 40 USD	80	11600	72.5
Delay Cost - promotion of 40 employees	14 days x 200 USD	2800	112000	
Total Estimated Cost	—	3080	133600	92.5



Conclusion:

- Time spent: 92.5 hours per reviewer, annually
- Manual and multi-step reviews: approximately USD 133,600 per year.

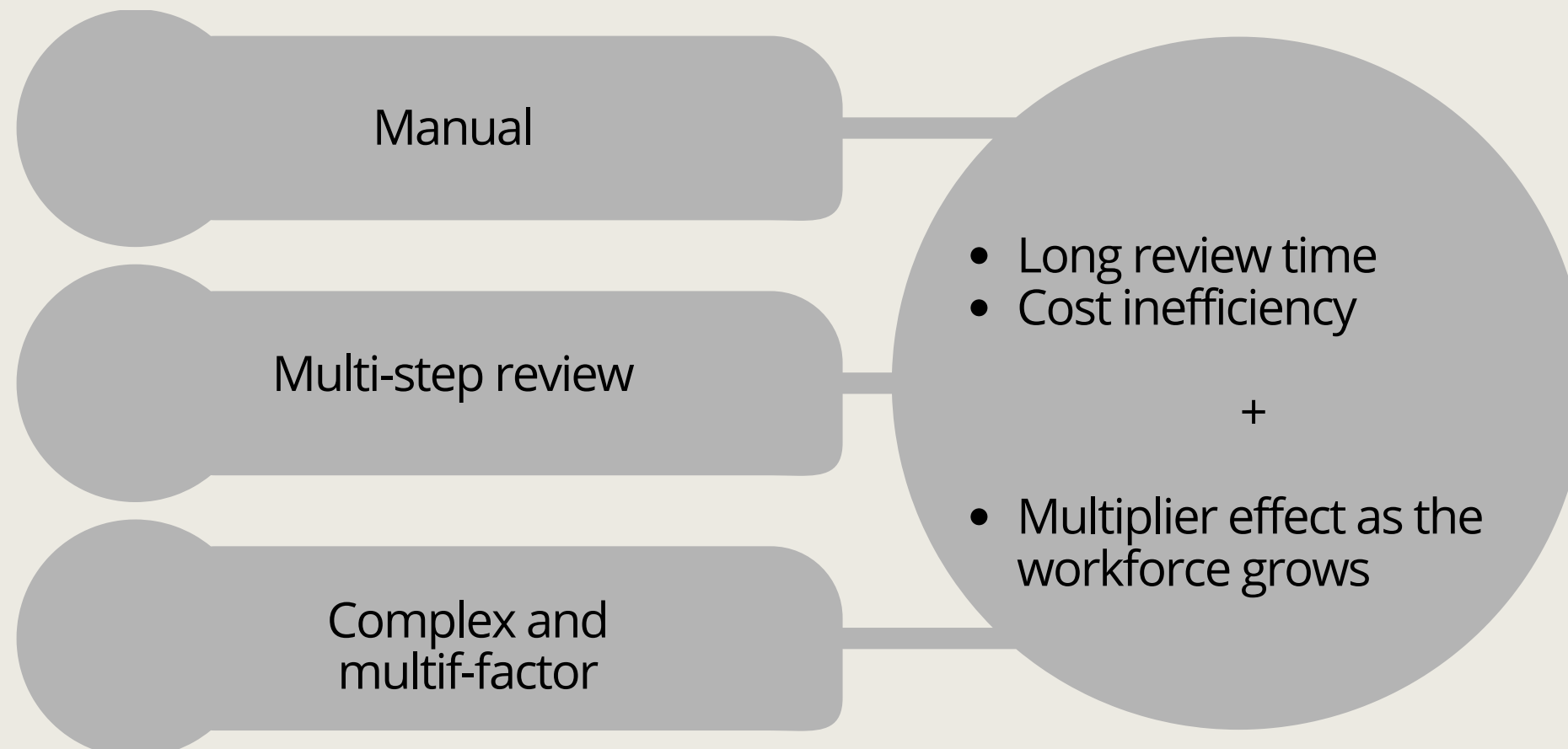
The manual process is repetitive and difficult to scale as the number of employees increases!

Problem Statement

The current employee promotion process is conducted through manual, multi-step reviews involving multiple stakeholders and various evaluation factors.

Due to its complexity, this process makes it difficult for HR to quickly prioritize high-potential candidates, delaying internal talent mobility and increasing the company's operational costs.

Employees Promotion Process



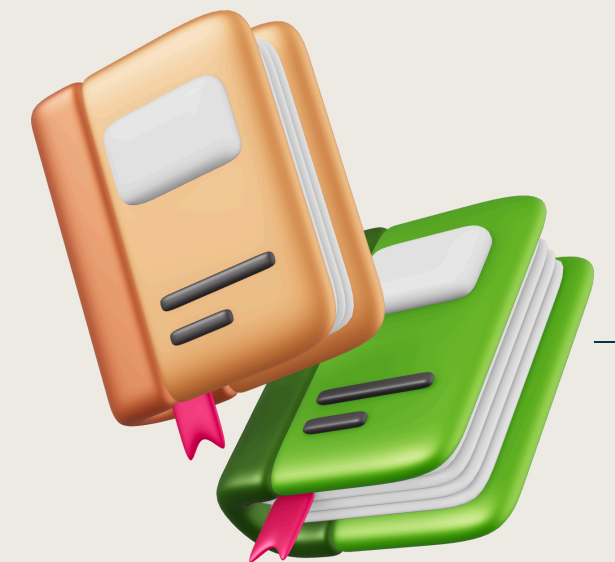
Goal

The goal of this project is to optimize the employee **promotion decision-making process** through a data-driven and automated approach.

By **leveraging machine learning**, HR can **save time and reduce operational inefficiencies**. The machine learning model serves to **predict promotion eligibility**, helping prioritize the most qualified candidates

"By automating manual and repetitive tasks, organizations can reduce administrative and operational costs by 30–50%."

(McKinsey, "Operations management reshaped by robotic automation", 2020)



OBJECTIVE, PROPOSED SOLUTION, & IMPACT

Potential Impact of the Automation Process as a Recommendation System

Objectives	Proposed Solution	Success Criteria	Reference
Reduce review time in employee promotion decisions	- Implement an ML model to generate eligible candidates based on performance, leadership, and project data. - Develop dashboard to provide insights displaying key indicators such as performance, leadership, and other relevant factors.	Review time and time-to-decision reduced by $\geq 30\%$	Deloitte (2023) – <i>Human Capital Trends Report</i>
Improve operational efficiency and reduce costs in the promotion decision process		Operational costs reduced by up to 30%	McKinsey (2021) – <i>The Future of HR: AI & Automation</i>

Machine Learning serves as a recommendation system in the initial screening stage, reducing screening time.

Machine Learning prioritizes candidates, decreasing review time by up to 30%.

Component	Formula	Cost/ Candidate (USD)	Total Cost (USD)
Initial screening – 500 employees, 1 manager per 50 employees - 30% less time - semiotomatis	10 mananger x 0.5 hours x 70% x 40 USD	60	3000
Formal review - 145 employees- 30%	4 reviewer x 0.5 hours x 70% x 40 USD	56	8120
Delay Cost - promotion of 40 employees - 30%	14 days x 70% x 200 USD	1960	78400
Total Estimated Cost	—	2076	89520
Saving		44080	32.99



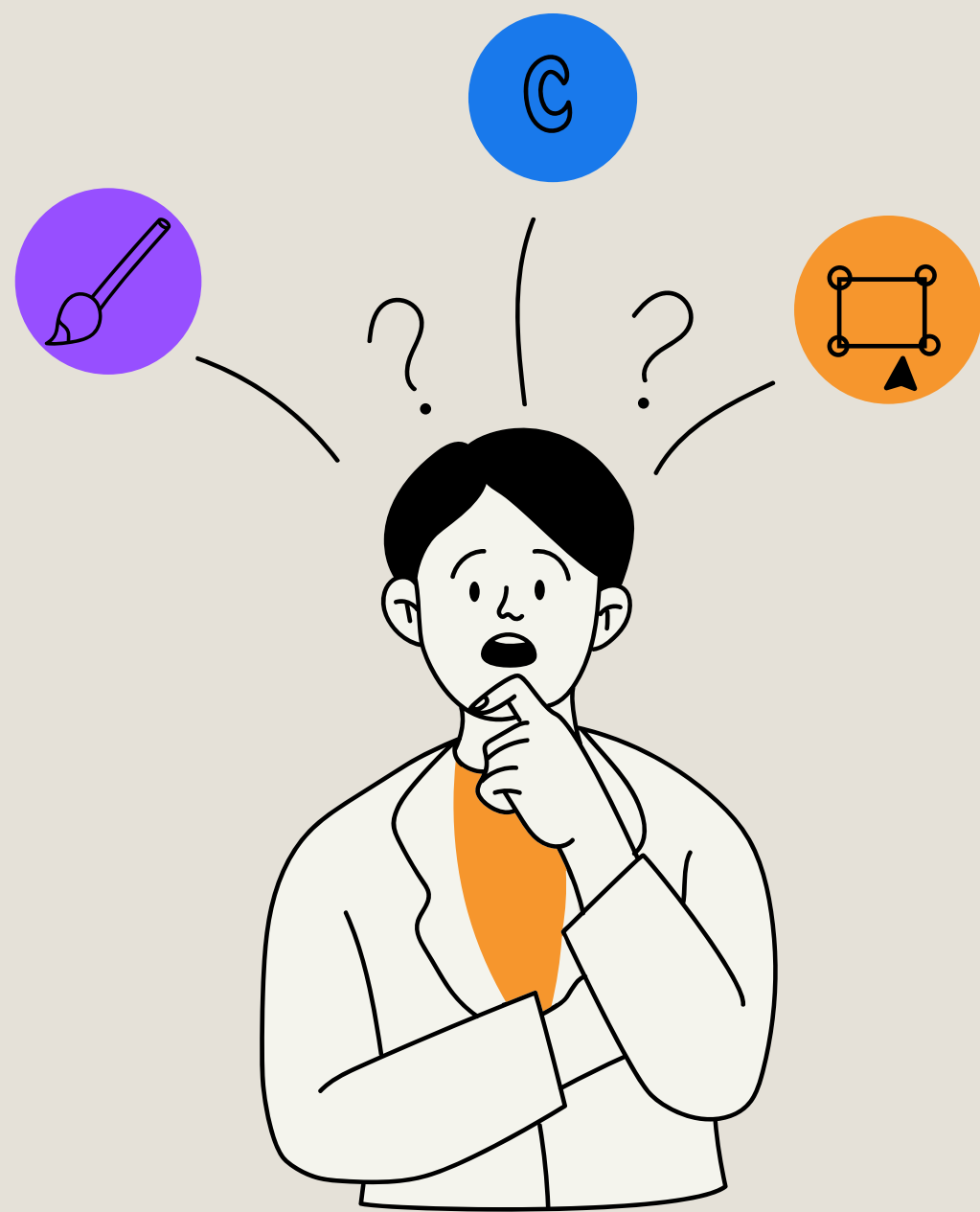
Potential savings:

USD 44,080/month ~ 33% compared to the manual process (USD 133,600)



DATA UNDERSTANDING

DATA EXPLORATION



Column	Description
Employee_ID	Unique ID for each employee
Age	Employee age
Years_at_Company	Years of service in the company
Performance_Score	Annual performance score (1–5)
Leadership_Score	Leadership score (0–100)
Training_Hours	Total training hours completed
Projects_Handled	Number of projects handled
Peer_Review_Score	Peer review score (0–100)
Current_Position_Level	Current job level (Junior, Mid, Senior, Lead)
Promotion_Eligible	Target label: 1 = eligible for promotion, 0 = not eligible

DATASET

Missing values :
@50 data in each column

Promotion_Eligible as target label

Anomaly values:
(-) Age and Years_at_Company

RangeIndex: 1000 entries, 0 to 999			
Data columns (total 10 columns):			
#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	Employee ID	1000 non-null	object
1	Age	950 non-null	float64
2	Years at Company	951 non-null	float64
3	Performance Score	950 non-null	float64
4	Leadership Score	950 non-null	float64
5	Training Hours	950 non-null	float64
6	Projects Handled	950 non-null	float64
7	Peer Review Score	950 non-null	float64
8	Current Position Level	950 non-null	object
9	Promotion_Eligible	950 non-null	float64

Numerical

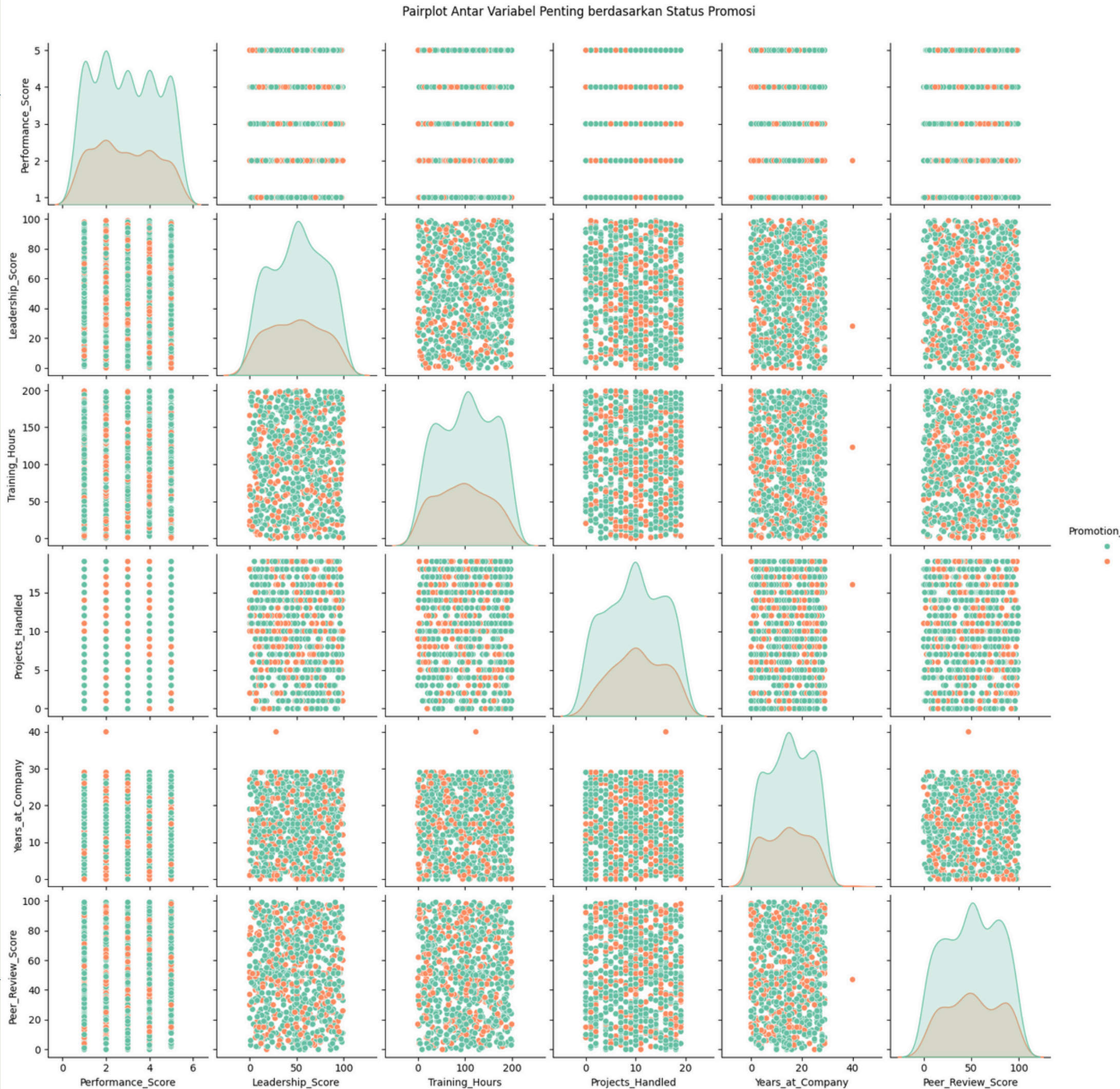
index	Age	Years_at_Company	Performance_Score	Leadership_Score	Training_Hours	Projects_Handled	Peer_Review_Score	Promotion_Eligible
count	9,500	9,510	9,500	9,500	9,500	9,500	9,500	9,500
mean	4,094	1,497	297	5,012	10,929	1,002	5,123	29
min	-30	-50	10	0	-500	0	0	0
medi	410	150	30	510	1,025	100	510	0
max	2,000	1,000	50	990	50,000	190	990	10

Categorical

	Employee_ID	Current_Position_Level
count	1000	950
unique	1000	4
top	EMP1000	Mid
freq	1	248

EXPLORATION DATA ANALYSIS

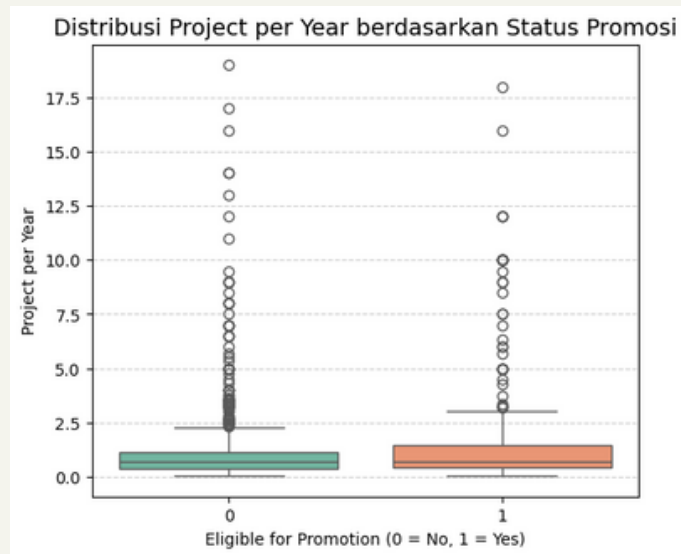
There is no apparent pattern of separation—indicating no strong relationship between the feature and Promotion Eligibility.



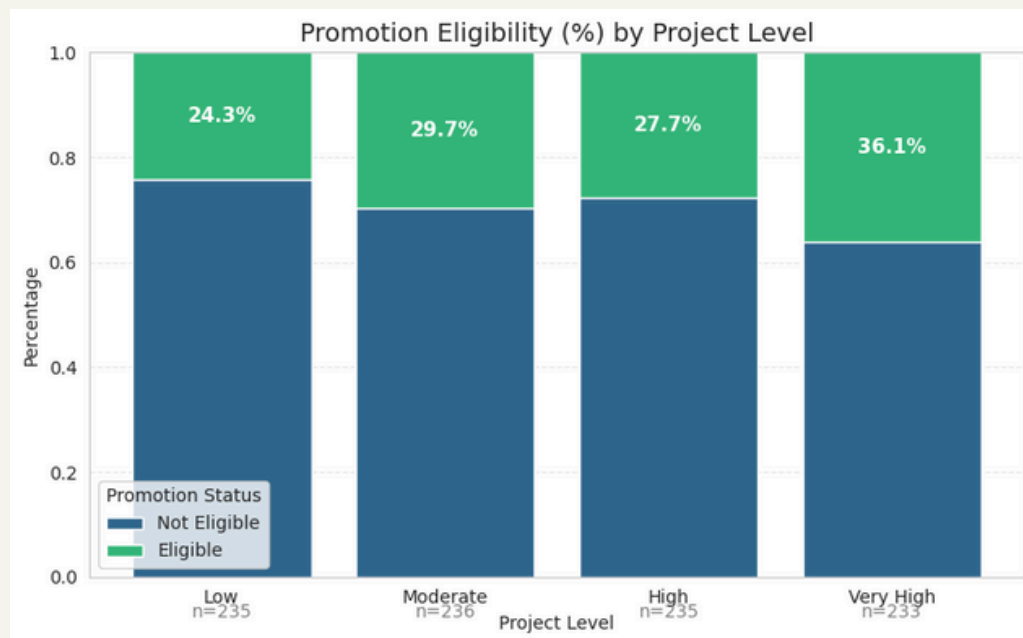
FEATURE ENGINEERING

PROJECT_LEVEL

= `pd.qcut (Project_Handled /
Years_at_Company`



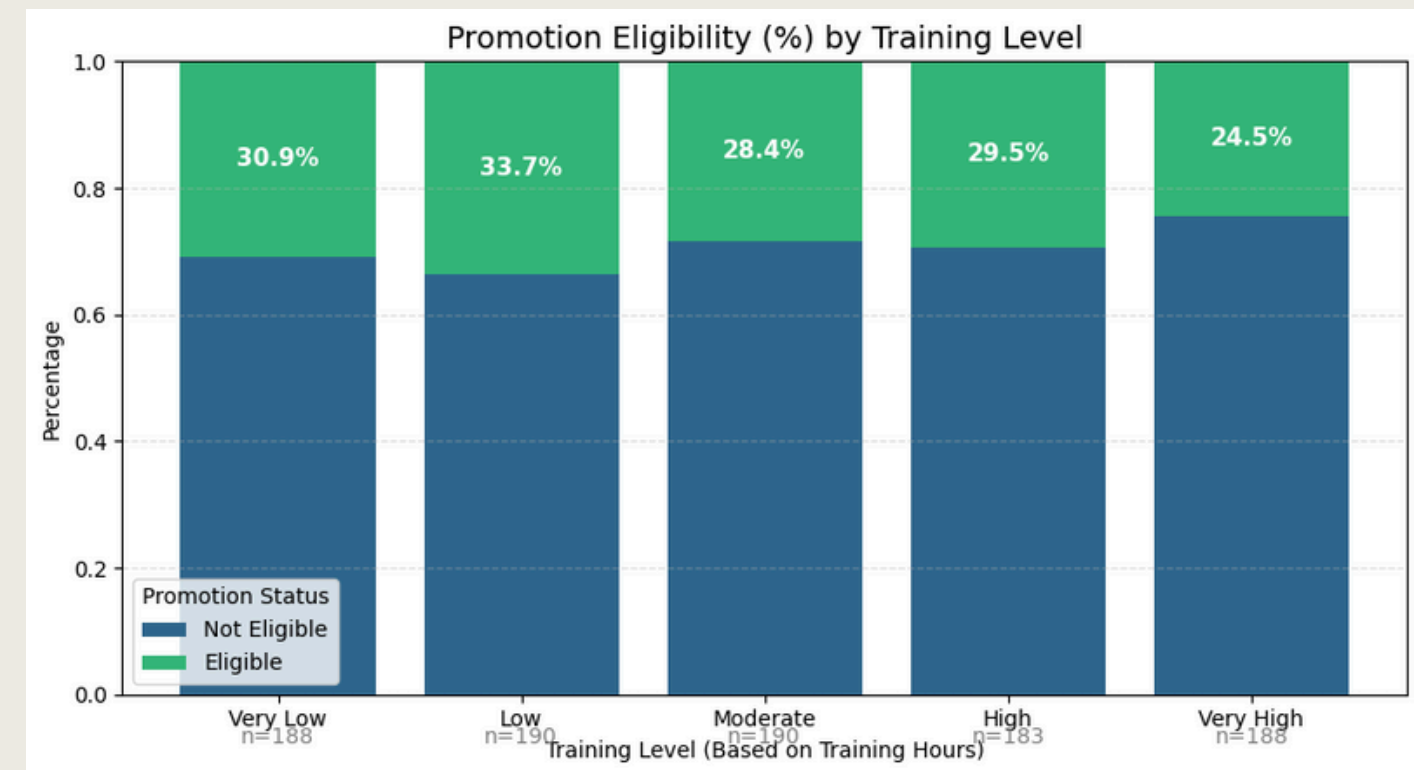
Visually, it appears that the productivity trend is slightly higher among promoted employees.



In the 'Very high' project level segment: the highest promotion percentage.

TRAINING_LEVEL

= `A pd.qcut (Training_Hours), q=5`



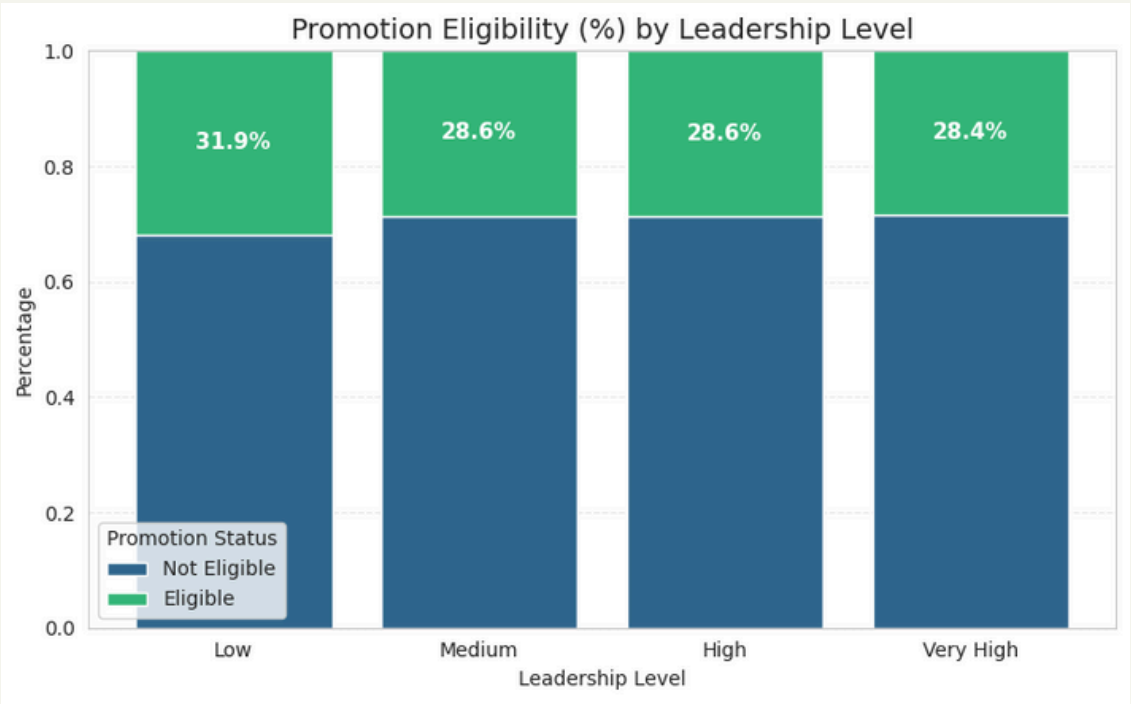
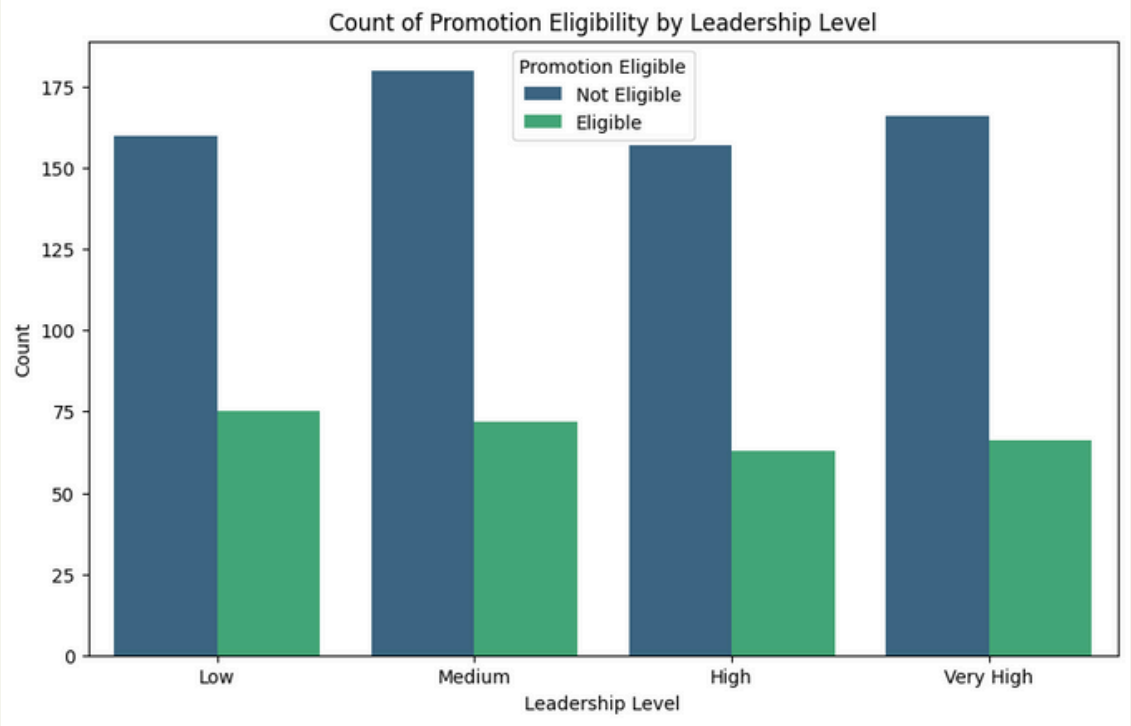
The second level (Low) training has the highest promotion eligibility.

FEATURE ENGINEERING

LEADERSHIP_LEVEL

=

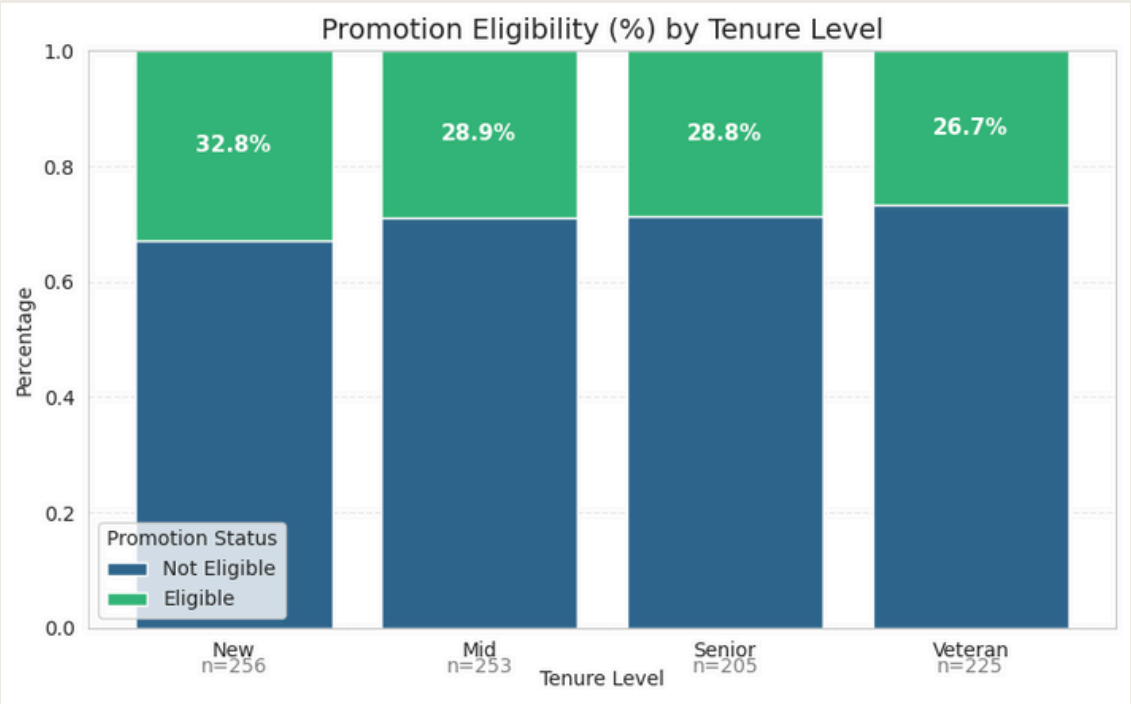
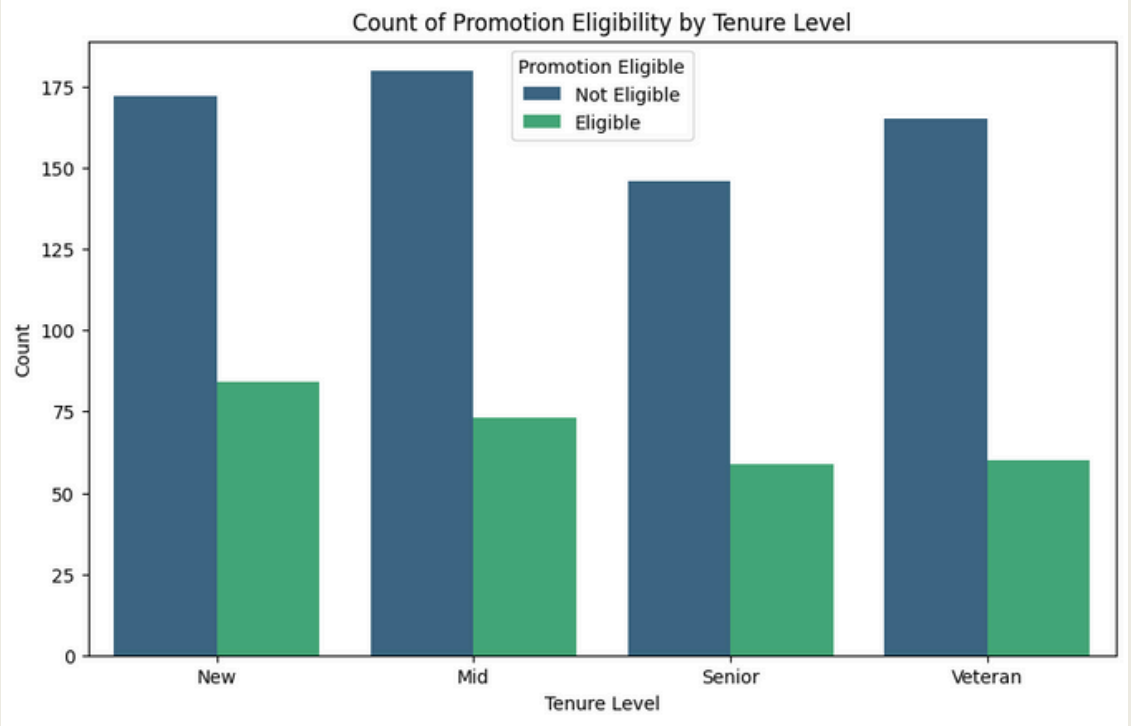
More employees at lower levels are promoted compared to higher levels.



TENURE_LEVEL

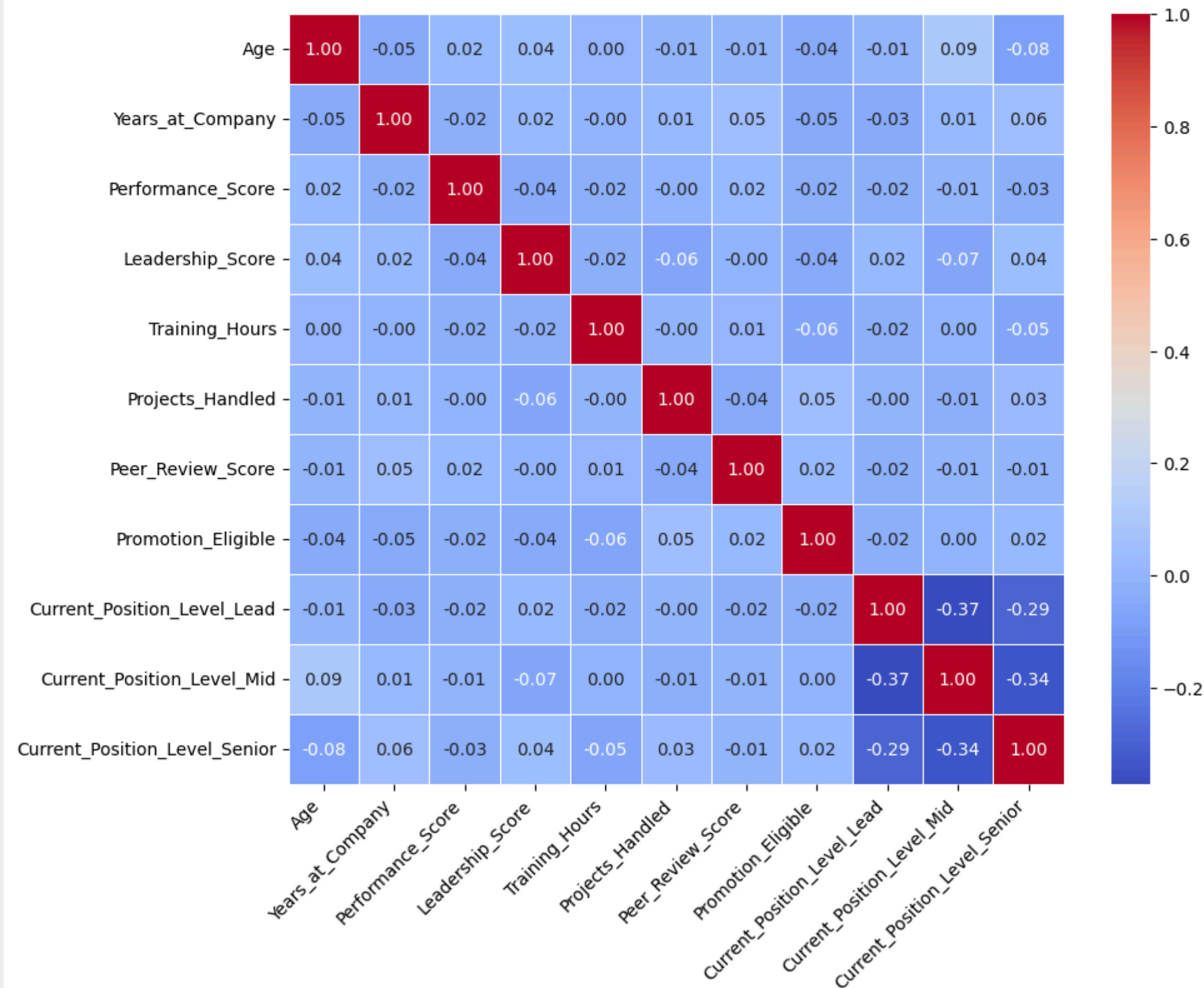
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The new employees are more eligible for promotion compared to the senior and veteran ones.



HEATMAP CORELATION

Heatmap Korelasi Fitur



INSIGHT:

- Feature correlations are weak → **no significant multicollinearity detected**
- Job level (Lead, Mid, Senior) shows a negative correlation – which is expected
- Age and tenure have a very weak relationship ($r = -0.05$), indicating independence
- Performance and leadership scores are independent
- **Promotion eligibility** is not linearly influenced by a single factor; it is likely driven by **non-linear or combinational effects**.



DATA PROCESSING

PRE-PROCESSING

Data Cleaning

Imputed missing values using mode or median/mean, removed extreme outliers using the IQR method, and converted Promotion_Eligible to integer format (0/1).

Feature Encoding

Applied One-Hot Encoding to Current_Position_Level to convert categorical data into numerical form.

Feature Engineering

Created new features — Project_Level, Training_Level, Leadership_Level, and Tenure_Level — to capture productivity and promotion readiness.

Feature Transformation & Standardization

Performed log transformation on skewed features (Age, Years_at_Company) and standardized scales using StandardScaler.

Handle Imbalance

Addressed class imbalance (29% eligible) using SMOTE and class_weight='balanced'.



MODELING

Metrics Selection:

- Accuracy : Provides an overview of the model's overall performance but is less sensitive to the minority class.
- ROC-AUC: Measures the model's ability to distinguish between positive and negative classes overall.
- PR-AUC: Focuses on the model's capability to detect eligible candidates; more suitable for imbalanced data (even after SMOTE balancing).
- CV PR-AUC: Evaluates the model's consistency across different data subsets during cross-validation.

MODEL

	Model	Accuracy_Train	Accuracy_Test	Precision_Test	Recall_Test	F1_Test	ROC_AUC_Test	PR_AUC_Test	CV_Recall_Test	CV_ROC_AUC_Test	CV_PR_AUC_Test	Comparison Result
1	LogReg Baseline	0.766	0.681	0.419	0.236	0.302	0.544	0.382	0.602	0.784	0.794	Fairly balanced
2	LogReg Tuned 2	0.76	0.676	0.406	0.236	0.299	0.541	0.376	0.611	0.787	0.796	Good enough, less significant tuning
3	XGB Baseline	1	0.633	0.348	0.291	0.317	0.508	0.344	0.675	0.814	0.817	Overfit
4	XGB Tuned	0.964	0.633	0.294	0.182	0.225	0.458	0.315	0.653	0.825	0.831	Overfit improved
5	DecisionTree Baseline	1	0.585	0.317	0.364	0.339	0.52	0.302	0.664	0.673	0.62	Overfit
6	DecisionTree Tuned	0.675	0.559	0.311	0.418	0.357	0.543	0.351	0.611	0.658	0.647	Overfit improved
7	RandomForest Baseline	1	0.681	0.4	0.182	0.25	0.535	0.347	0.696	0.854	0.864	Overfit
8	RandomForest Tuned	0.948	0.707	0.5	0.164	0.247	0.506	0.347	0.657	0.83	0.837	Highest accuracy, low recall, but ROC-AUC and PR-AUC are sufficient
9	AdaBoost Baseline	0.748	0.649	0.333	0.2	0.25	0.535	0.363	0.619	0.774	0.78	Fairly balanced
10	AdaBoost Tuned	0.973	0.633	0.316	0.218	0.258	0.493	0.305	0.666	0.813	0.831	Fairly balanced, tuning accuracy drops
11	SVC Baseline	0.863	0.676	0.364	0.145	0.208	0.548	0.342	0.625	0.813	0.818	Fairly balanced
12	SVC Tuned	0.984	0.59	0.304	0.309	0.306	0.518	0.311	0.698	0.783	0.764	Fairly balanced & stable
13	KNN Base	0.88	0.628	0.364	0.364	0.364	0.584	0.338	0.777	0.826	0.79	Fairly balanced & stable
14	KNN Tuned	0.882	0.622	0.352	0.345	0.349	0.619	0.357	0.781	0.838	0.804	Fairly balanced & stable, accuracy drop

SELECTED MODEL

Model:

Random Forest - tuned

Reasoning:

- Highest test accuracy (0.707)
- Highest test precision (0.50)
- Stable ROC-AUC & PR-AUC scores (≈ 0.50 / 0.35)
- Most robust against overfitting

Kelemahan:

- Recall remains low (0.16) — requires threshold tuning to increase sensitivity toward the “positive” class

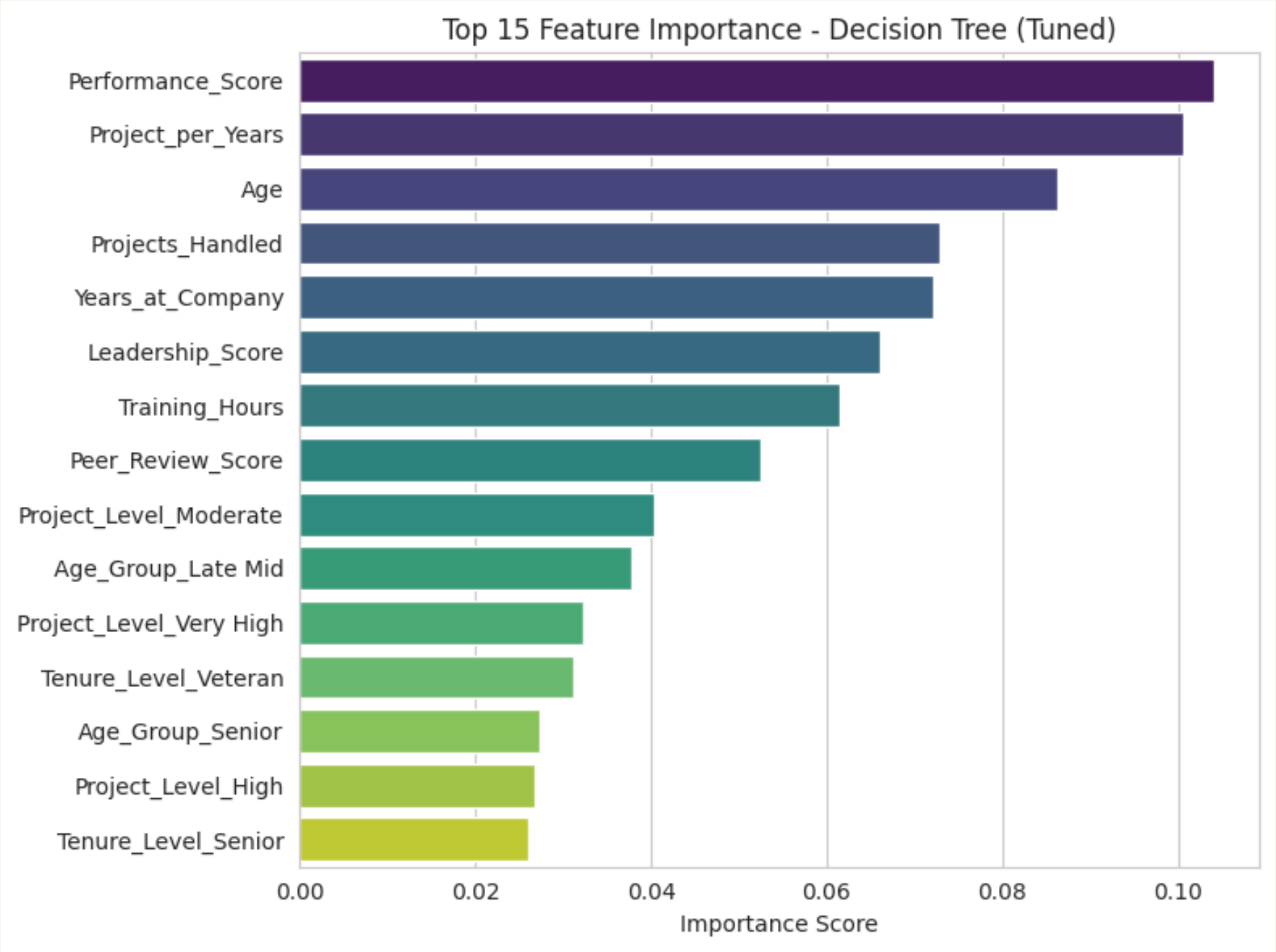


FEATURE IMPORTANCE

THE MOST DOMINANT
FACTOR

FAIRLY DOMINANT
FACTOR

MINOR INFLUENCING
FACTOR



THE MOST DOMINANT
FACTOR

- Performance Score
- Projects per years
- Age

FAIRLY DOMINANT FACTOR

- Project Handled
- Years at Company
- Leadership Score
- Training Hours

MINOR INFLUENCING FACTOR

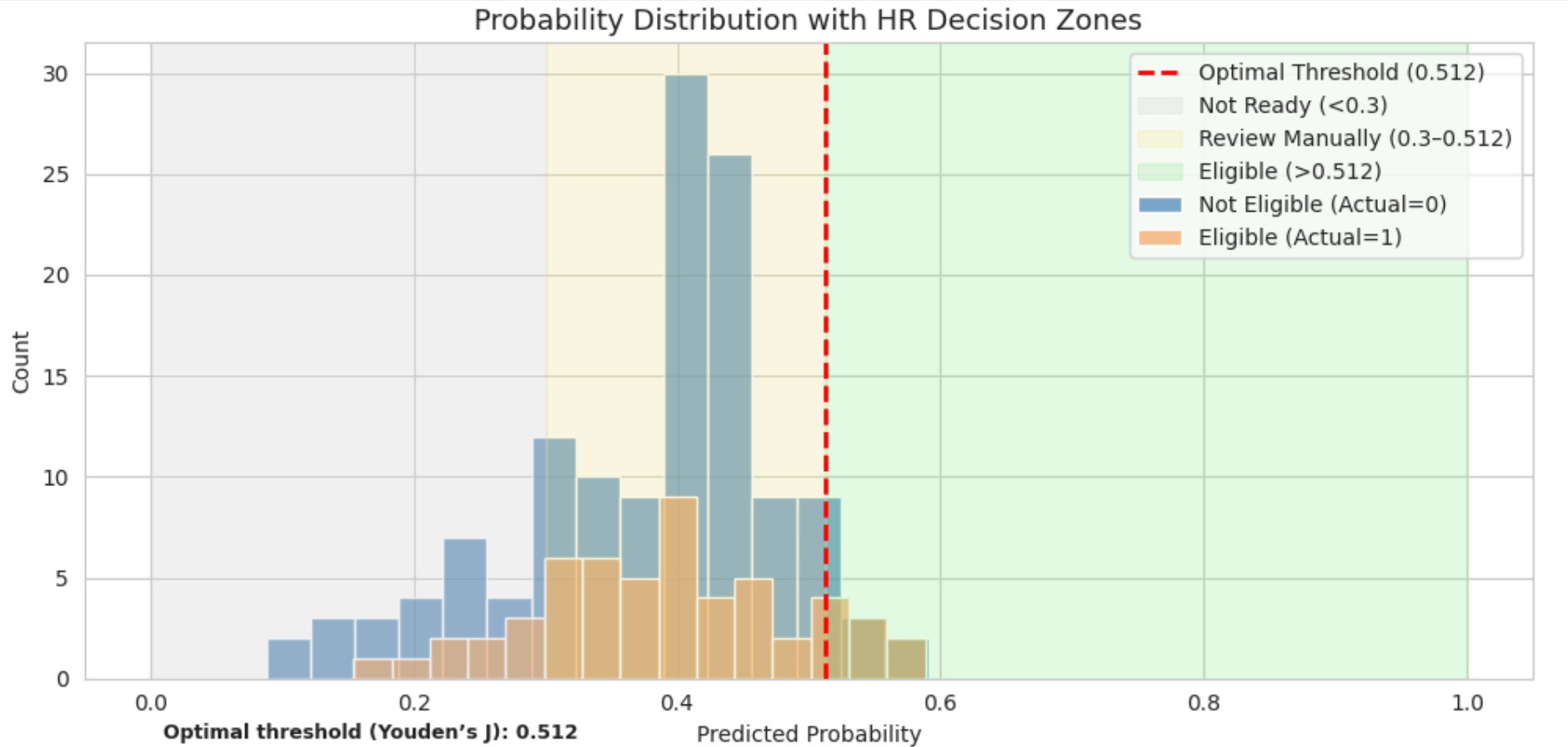
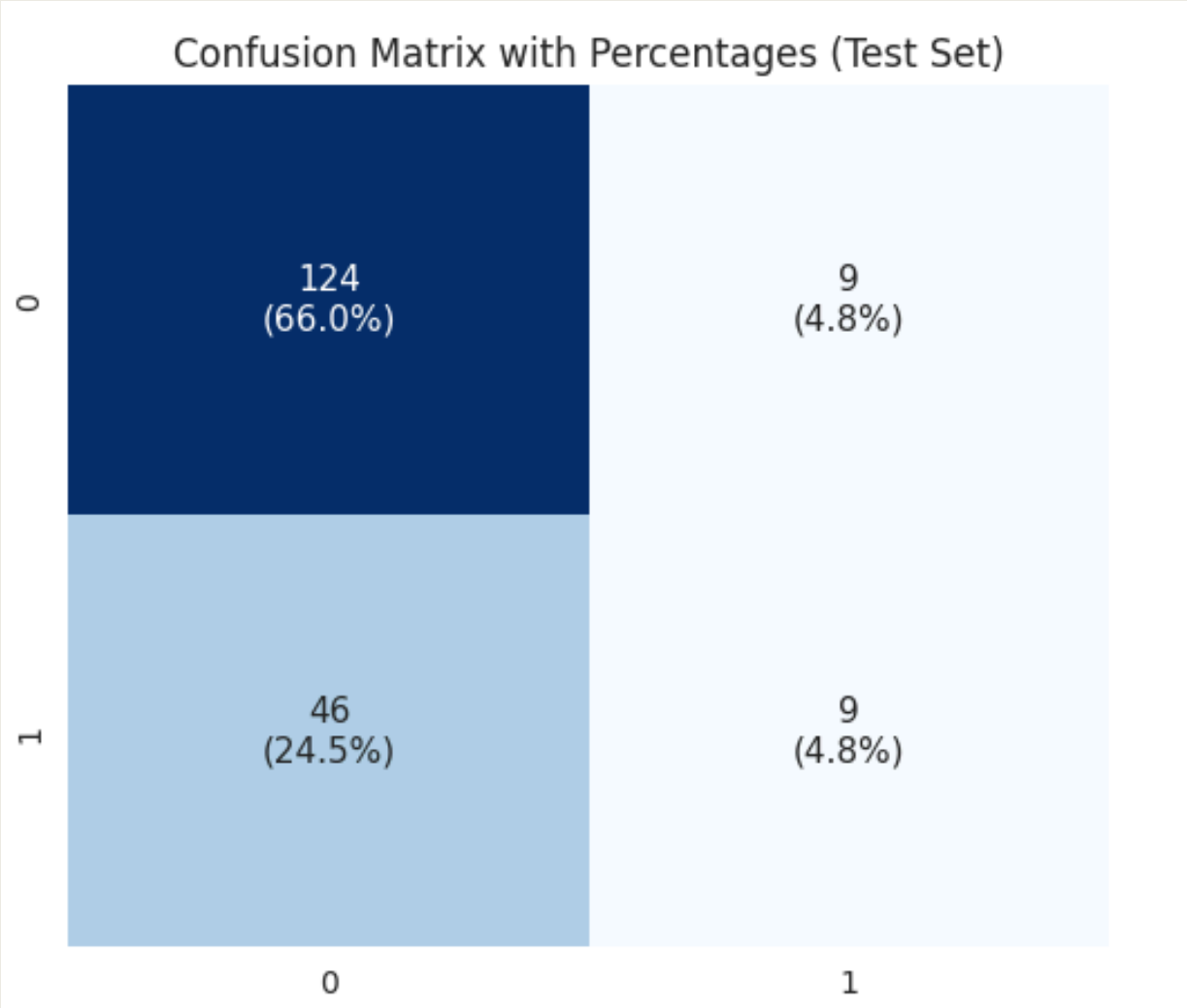
- Peer review
- Project level moderate
- Age group late mid



MODEL INTERPRETATION

FEATURE	COLOR AND POSITION PATTERN	INTERPRETATION
Project Handled	All points cluster on the left; red points slightly shift to the right	Although the total number of projects shows a generally negative impact, having more projects is still better than having fewer.
Projects_per years	Red points tend to spread toward the right	Higher project intensity contributes positively to promotion probability.
Project_level_moderate	Red points are concentrated on the right	Moderate project productivity shows a positive contribution to promotion probability.

EVALUATION



Error Type	Percentage	Assumption for 500 employees	Category
TN	66%	330 employees	Not ready
TP	24.4%	122 employees	Confirmed Eligible
FN	4.8%	24 employees	Missed potential
FP	4.8%	24 employees	Overestimated

Out of 500 employees, the model correctly identifies 330 as not yet promotion-ready and 122 as eligible for promotion, with about 24 missed and 24 wrongly suggested or overestimated.

- Insights:
- Model highlights clear promotion potential zones — green area shows high-confidence candidates ready for promotion.
 - Most employees fall in mid to low range, indicating opportunity for targeted development.
 - Manual review remains essential for mid-range scores to ensure no high-potential employees are overlooked.



BUSINESS IMPACT & RECOMMENDATION

BUSINESS IMPACT

The following are the business impacts of the model:

Aspect	Interpretation
Overall Accuracy	The model correctly predicts about 71% of the test data (Test Accuracy = 0.707).
Main Error	The model often fails to identify employees eligible for promotion (Low Recall: 16%) compared to incorrectly promoting non-eligible ones (Precision: 50%). This indicates a high false negative rate and a lower false positive rate.
HR Implication	Many high-potential talents may be overlooked due to the high false negative rate → manual review or a model with higher recall is needed.
Recommendation	The model should be used as an initial screening tool rather than a final decision-maker

BUSINESS IMPACT

The following classification helps HR prioritize promotion reviews and identify cases that still need manual evaluation :

Error Type	% of Employees	Est. Count (of 500)	Probability Range	Category	Interpretation	Key Action for HR
TP	24.40%	122	> 0.70	Confirmed Eligible	Strong confidence – high likelihood of promotion readiness	Proceed to promotion review / fast-track
FN	4.80%	24	0.40 – 0.50 (below threshold)	Missed Potential	Model missed some actual high performers near the threshold	Manual review – may be strong performers
FP	4.80%	24	0.50 – 0.65 (above threshold)	Overestimated	Predicted eligible, but actual data does not support it	Validate data completeness / performance records
TN	66%	330	< 0.40	Not Ready	Consistently low probability – unlikely promotion-ready	No immediate action needed

Output example :

Employee_ID	Probability 	Predicted 	Category 	HR_Action 
128	0.418383	0	Not Ready	No immediate action needed
38	0.449893	0	Missed Potential	Manual review – may be strong performer
151	0.453461	0	Not Ready	No immediate action needed
483	0.318358	0	Not Ready	No immediate action needed
115	0.406744	0	Not Ready	No immediate action needed

BUSINESS IMPACT

Direct Review Reduction (Time/Cost):

The model still helps streamline the review process by reducing the number of candidates requiring manual evaluation, despite its relatively low recall rate.

Component	Formula	Cost/ Candidate (USD)	Total Cost (USD)	Total Time (hour/reviewer)
Initial screening - ML assisted, manager quick scan	10 manager x 0.5 hour x 40 USD	0	200	0.5
Formal Review - Eligible 122 candidates	4 reviewer 0.5 hour x 40 USD	80	9760	61
Formal Review - Missed Potential 24 candidates	4 reviewer x 0.5 hour x 40 USD	80	1920	12
Formal Review - Overestimated 24 candidates	4 reviewer x 0.30 hour x 40 USD	48	1152	12
Delay cost - ML, Missed Potential 24 candidates	14 days x 200 USD	2800	67200	
Total Estimated Cost		160	80232	73.5
Saving %	Total cost manual = USD 133 600; Total time manual = 92.5 hrs		39.95	20.54

This represents a savings of approximately ~40.0% per annum and administrative time reduced by 20% for a company with 500 employees.

Note: The savings would be more optimal with a higher recall. The current low recall keeps the delay cost high, as manual review is still required to identify the qualified candidates missed by the model.

RECOMENDATION

Short-term Recommendations

1. Optimization of Model & Data Processing

- Improve preprocessing by addressing missing values-handling outliers or different data imbalance handling.
- Fine-tune the model to enhance recall, for example, through threshold adjustments or weighted loss techniques.

2. Implement the model as a recommendation system

Prioritize reviewing employees not recommended by the model based on the prediction probability ranking for each employee.

Long-term Recommendations

Additional Features and Existing Feature Audit

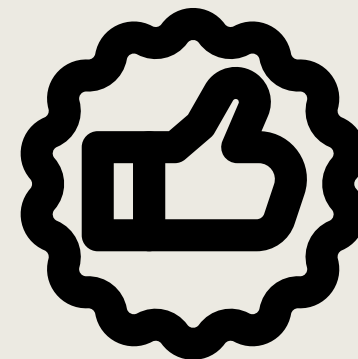
- Soft-skill/leadership behavior metrics (e.g., 360 feedback, peer review insights)
- Performance score review/audit

Project complexity/impact/quality → not just the number of projects



Model Optimization & Data Processing

- Improve preprocessing
- Tune the model to improve recall, for example through threshold adjustment or weighted loss.



Recommend Model as a Scoring Tool:

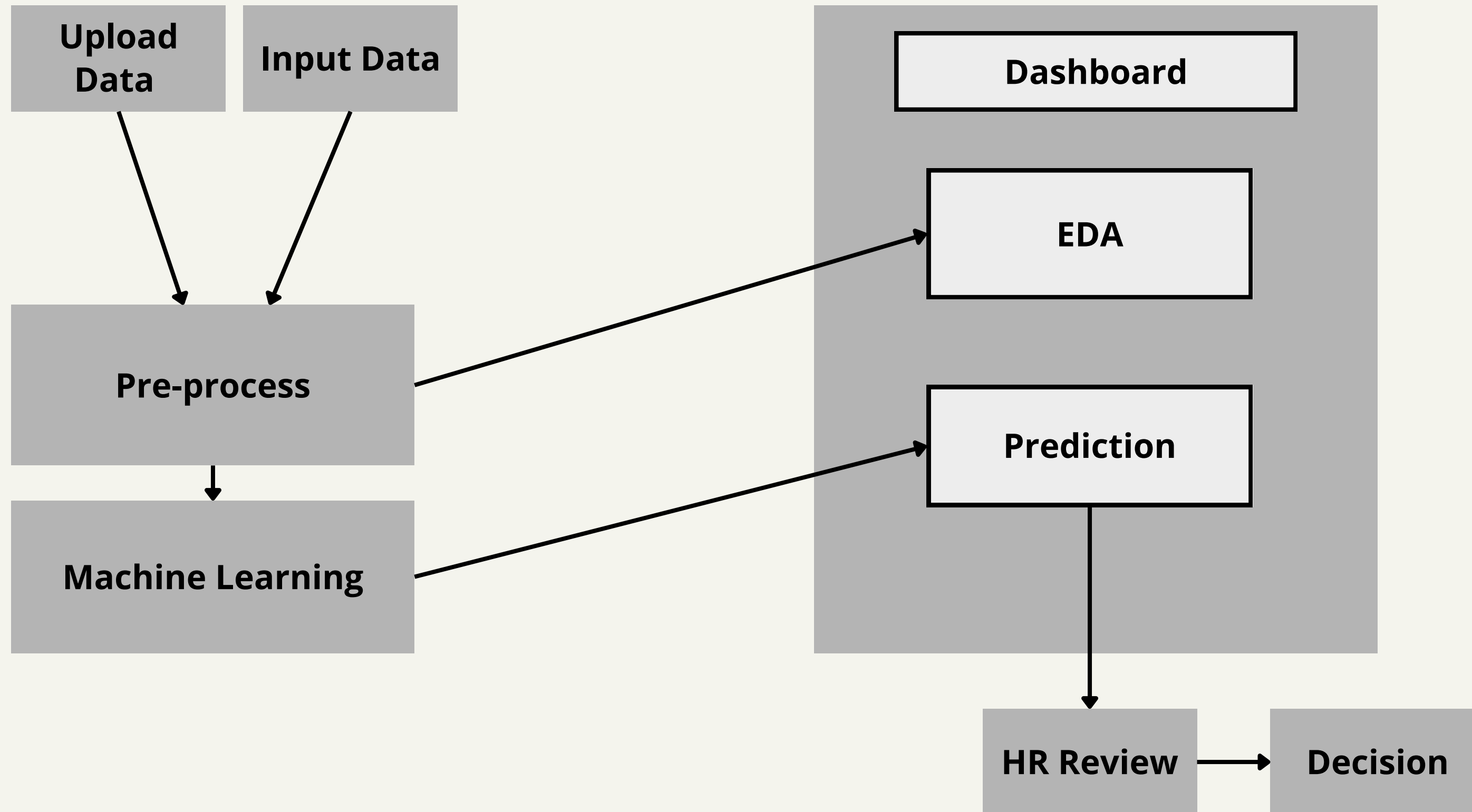
Once the model's performance is improved (Recall > 50%), it should be used solely as a scoring tool (e.g., providing a score from 1–100) rather than as an automatic “Yes/No” decision-making system.



Criteria Audit: Add or refine features :

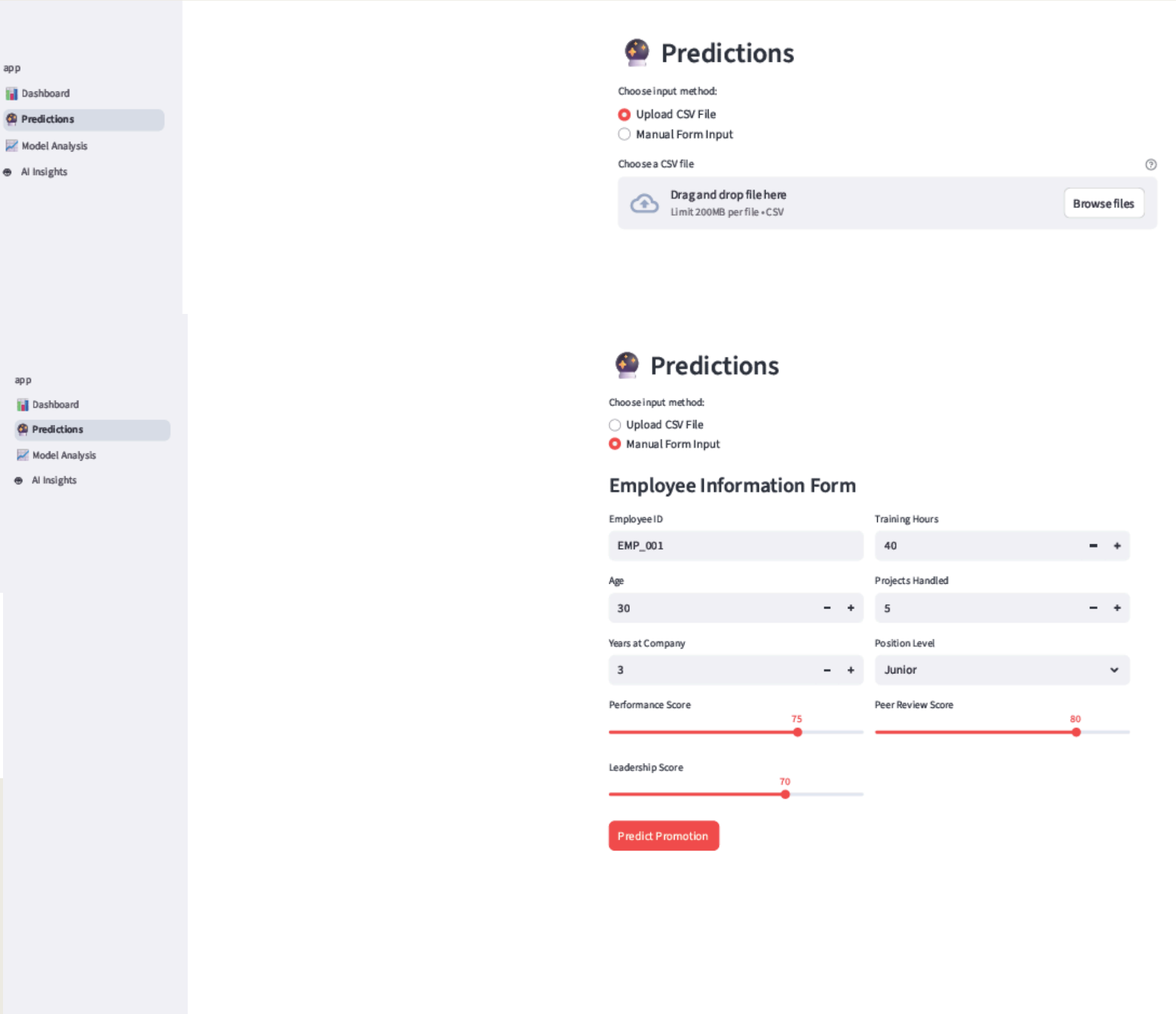
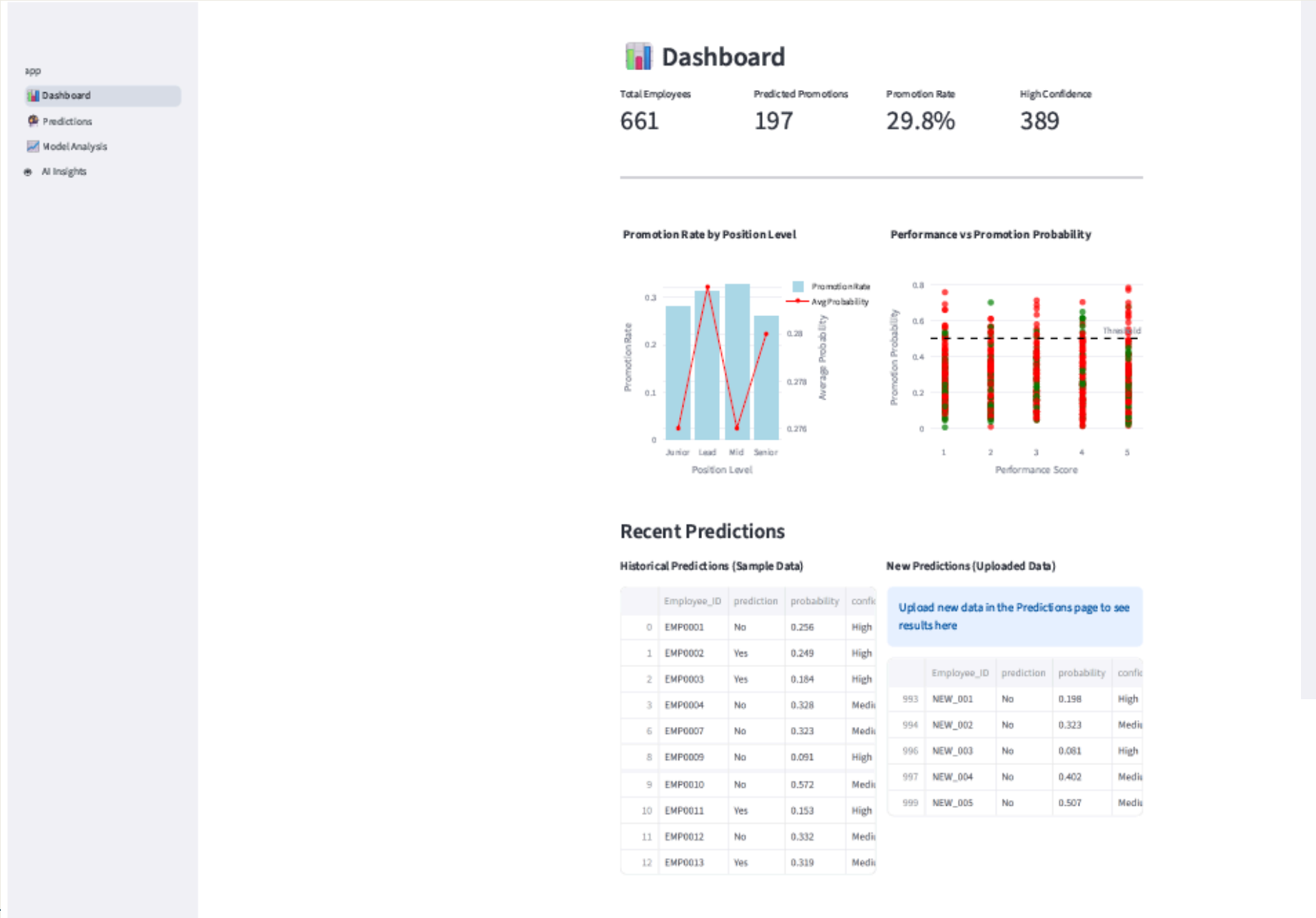
such as soft-skill or leadership behavior metrics (e.g., 360 feedback), project complexity/impact, and review whether current performance metrics truly reflect promotion criteria.

DEPLOYMENT



DEPLOYMENT: STREAMLIT APP

Link



CONCLUSION

The employee promotion recommendation system project, powered by machine learning, achieved an optimal model using **Random Forest**. With an accuracy of **71%** and a precision of **50%**, the model effectively highlights high-potential candidates based on performance, leadership, and project scores.

This recommendation system enables HR to focus on the most promising employees, improving promotion process efficiency while reducing time-workload and cost by approximately **40% per year**.



THANK YOU